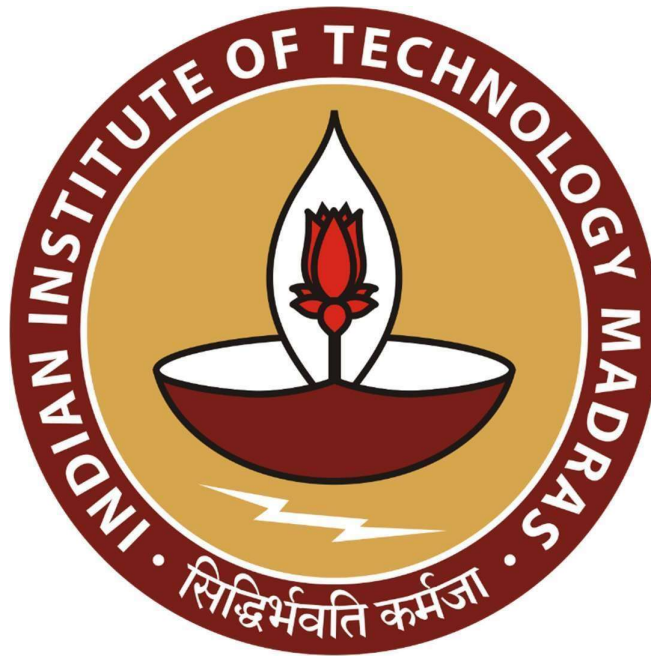


DATA DRIVEN ANALYSIS OF POULTRY FARM OPERATIONS AND LIVE CHICKEN SUPPLY

An End-Term report for the BDM capstone Project

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1. Executive Summary

This study focuses on Karthik Reddy Poultry, a small-scale poultry farm located in Siddipet district, Telangana. The farm supplies live chickens to nearby meat shops and local businesses and operates with multiple batches throughout the production cycle. However, it faces several operational challenges such as high chick mortality, inconsistent sales patterns, and increasing inventory costs related to feed and medicines. These issues directly affect the farm's productivity, profitability, and overall operational efficiency, making it necessary to analyze the underlying factors using a data-driven approach.

The data for this study was collected over a three-month period from informal farm records and observations. The dataset includes daily chick mortality details, environmental conditions such as shed temperature, vaccination status, and litter condition, along with sales transactions and inventory consumption data. Descriptive statistical measures were used to understand the variability and patterns in the data. Multiple linear regression was applied to analyze the combined impact of environmental factors on mortality. Time-series analysis was used to study sales trends over time, and ABC analysis was performed to classify inventory items based on their consumption value and importance.

The results indicate that chick mortality is influenced by multiple factors, with temperature showing a relatively stronger relationship compared to other variables, while vaccination and litter condition show moderate effects. The combined regression model provides a better overall understanding of mortality patterns compared to individual analysis. Sales data shows a gradual increase in demand over time with some fluctuations, indicating improving business performance. Inventory analysis highlights that a small number of items contribute to the majority of total cost. Based on these findings, maintaining optimal temperature, ensuring timely vaccination, and improving litter management can help reduce mortality. Additionally, better planning of high-cost inventory items and monitoring sales trends can improve efficiency, reduce losses, and support better decision-making in the farm.

2. Detailed Explanation of Analysis Process/Method

2.1 Data Collection

The raw data for this project was collected from Karthik Reddy Poultry farm located in Chinna Masanpally village, Siddipet district, Telangana. Primary data was directly obtained from the owner through daily operational records, manual registers, purchase bills, and sales records maintained between July and September 2025.

The collected data included chick mortality details, environmental conditions such as shed temperature, and humidity, sales transactions and inventory consumption records. Key Variables such as chicks died per day, mortality rate, shed temperature, humidity, birds sold, total sales amount, consumed quantity of feed and medicines, and total consumption value of inventory items.

The collected datasets are organized into three main categories: chick mortality data, sales data, and inventory data. These datasets form the foundation for statistical analysis of poultry farm's operational performance.

2.2 Data Cleaning and Pre-Processing Explanation

The raw data obtained poultry had contained minor inconsistencies such as missing values, outliers, inconsistent data formats, and variations in item naming. The data was cleaned and standardized before performing statistical analysis.

The dataset was carefully cleaned and prepared before analysis. Duplicated records were identified and removed to maintain data accuracy. Missing values were handled in spreadsheets by verifying details with farm owner whenever necessary. Item names and numerical values were normalized to avoid inconsistencies, and quantities were converted into normal measurable units. Additional variables, including total consumption value and mortality rate, were also made for deeper analysis.

Importance

Data Cleaning is an important step to ensure reliability and accuracy of analysis. Clean and Structured data reduces errors and helps in obtaining meaningful insights. Proper preprocessing ensures that regression analysis, time series analysis, and ABC classification produce better and good results. Without proper cleaning, the analysis could cause incorrect conclusions and ineffective business results. Data quality is essential for better conclusions from the data.

A sample of the cleaned dataset is provided in the spreadsheet link: [Dataset Link](#)

2.3 Descriptive Statistical Analysis

Descriptive Statistics was performed to understand the overall behavior and variability in check mortality, sales performance, and inventory consumption. Statistical measures such as mean, median, minimum, maximum, standard deviation, and quartiles were calculated for relevant variables.

The mean and standard deviation is calculated using

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

x_i : represents each individual data value

n : represents the total number of observations

$\bar{x}$: represents the mean (average) value of the dataset

Σ : represents the summation of all values

Descriptive Statistics of Sales data: [Descriptive Statistics of Birds Sold and Total Sales Amount](#)

- The average daily birds sold is around 33 birds, indicating steady demand in the poultry business.
- The total sales amount shows proper variation between minimum and maximum values, reflecting fluctuations in revenue.
- Standard deviation in both birds sold and sales amount suggest moderate variability in demand and earnings.
- Distribution of sales data show gradual improvement over time with some peak demand days.

Justification:

The sales descriptive statistics gives an overall understanding of sales and revenue patterns in Karthik Reddy farm and helps in fluctuations in daily business performance.

Statistic	Chicks_Died_Today	Mortality_Rate
Count	92	92
Mean	1.656	0.626
Median	1.35	0.44
Minimum	0	0
Maximum	4	1.82
25%	1	0.34
50%	2	0.67
75%	2	0.885
Standard deviation	1.133	0.455

Fig-1. Descriptive statistics of Chick mortality rate and Chick deaths in a day

Chicks_Died_today represents the number of chicks that died on a particular day

- Total 92 days of mortality data was analyzed from the farm.
- The average chick mortality rate is around 1.65, indicating regular mortality in batches.
- Average mortality is around 0.626, showing loss affecting productivity.

Justification:

This mortality descriptive statistics helps to understand mortality pattern and variation in the poultry farm. This gives clear base to analyze relationship between environmental factors and mortality using regression analysis and supports identifying the main cause of chick loss.

Descriptive statistics of Inventory Data: [Descriptive Statistics of Inventory Consumption \(Quantity and Value\)](#)

- Broiler Feed has the highest average consumption among all items.
- Starter feed is the second most consumed item, showing its importance in early growth.
- Vaccines and vitamins have lower quantities but are essential for bird health.
- Hygiene items such as disinfectant and litter material show neutral usage.

Justification:

Based on item-wise consumption pattern of all inventory materials used in the poultry farm, ABC analysis is formed.

- Class A items are Broiler Feed and Starter Feed
- Class B items are Vaccines and Vitamins
- Class C items are Disinfectant and Litter Material shows lesser significance.

ABC classification helps the farm focus more on high-cost items, improve inventory planning, and reduce unnecessary expenses.

2.4 Regression Analysis for Chick Mortality

Explanation:

Multiple Linear regression analysis is used to examine the relationship between chick mortality rate and environmental as well as operational factors. The dependent variable used in regression model is mortality rate. Shed temperature, humidity, Ventilation status, and litter condition.

Regression model is simple curve fitting.

Mortality Rate = $\beta_0 + \beta_1(\text{Temperature}) + \beta_2(\text{Humidity}) + \beta_3(\text{Vaccination}) + \beta_4(\text{Ventilation}) + \beta_5(\text{Hygiene}) + \epsilon$

Where $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ represents regression coefficients and ϵ is random error.

The regression model is developed using python and Excel tools to measure the impact of each factor on chick mortality and to ensure conclusions to reduce revenue losses.

Variable	Coefficient	Impact on Mortality
Temperature	0.1147808995	Increases Mortality
Humidity	0.05516855235	Affects mortality
Ventilation	-0.4992649939	Reduces mortality
Hygiene	-0.4441385922	Improves survival
Vaccination	-0.00186733958	Reduces mortality
Intercept	-3.004175765	base mortality level
R2 Score	0.815	coefficient of Determination

Fig-2: Multiple Regression Analysis of Factors affecting Chick Mortality Rate

Key Observations from Regression Analysis

- Regression analysis was conducted to examine how environmental and management factors influence chick mortality rate.
- The coefficient for shed temperature (0.1148) indicates a positive relationship with mortality, meaning higher temperature leads to increased chick deaths.
- Humidity also shows a positive coefficient 0.0551, suggesting that rising humidity levels slightly increase mortality rate.
- Ventilation has a negative coefficient -0.4992, showing that proper ventilation helps reduce chick mortality.
- Hygiene condition (litter condition) also has a negative coefficient -0.4441, indicating that better hygiene and dry litter improve chick survival.
- The intercept value -3.0041 represents the base mortality level when all independent variables remain constant.
- Ventilation has negative coefficient of value -0.00186 indicates proper vaccination to chicks reduce chick death in the poultry farm.
- The R^2 value of 0.815 shows that about 81.5% of variation in mortality rate is

explained by the selected variables, indicating a strong model fit.

Justification:

The Regression analysis helps identify the major factors responsible for chick mortality in the poultry farm. By maintaining proper temperature, humidity control, ventilation, and hygiene, the farm can significantly reduce mortality and improve overall productivity.

2.4 Time Series Analysis (Sales Data)

Explanation:

Time Series is used for monthly and daily sales data to understand revenue patterns and demand fluctuations over time. Sales amount and number of birds sold were analyzed using daily farm records. Line charts and monthly aggregation are used to observe trends and to understand revenue stability and demand growth across the 3 month period.

Time series analysis is applied to poultry sales data which is arranged in Chronological order. Time series analysis helps in identifying increasing or decreasing sales trends and seasonal variations across months.

Time series focuses on:

$Y_t = \text{Trend} + \text{Seasonal Component} + \text{Random Error}$ Where $Y_t = \text{Sales}$

at time t

Month	Total Sales	Birds Sold
July	134760	668
August	264209	1003
September	456521	1338

Fig-3: Month-wise Sales and Demand Summary

Birds Sold represents the number of chickens sold, while Total Sales Amount represents the total revenue generated from those sales.

- Sales show a steady increase from July to September, indicating improving business performance.
- September recorded the highest sales and demand, showing strong market response.
- Bird sales also increased each month, reflecting rising customer demand.
- Growth in both sales and quantity sold indicates positive business expansion.

Month	Avg Day Sales
July	4347.096774
August	8522.870968
September	15217.36667

Fig-4: Average Daily Sales Trend

- Sales improved significantly in August as more birds reached market weight and demand increased.
- September shows the highest average daily sales due to higher bird availability and consistent buyer demand from meat shops and hotels.
- Since the farm supplies birds in bulk to multiple buyers, daily sales value depends on number of birds sold and market price, resulting in higher revenue during peak selling periods.
- The increase in sales during September can also be linked to the festive season (Dussehra period), during which demand for non-vegetarian products generally rises, leading to higher poultry sales and revenue.

Justification:

The poultry farm experiences fluctuations in sales and revenue, which affects financial Stability. Time-Series analysis helps to identify sales growth pattern and demand

variations. Understanding these patterns better production planning, pricing decisions, and batch scheduling. Therefore, this method directly addresses the issue of inconsistent Sales and revenue fluctuations.

2.5 ABC Analysis for Inventory Prioritization

Explanation:

ABC analysis was applied to classify inventory items based on their total consumption value. Consumption value was calculated using the formula:

Consumption Value= Quantity Consumed x Unit Cost

Items were sorted in descending order of the consumption value. All Inventory items such as feed, medicines, and disinfectants are arranged in descending order of Consumption value. Based on cumulative percentage, items are classified in three categories

- Class A: High value items contributing to major cost (70-80% items)
- Class B: Medium Value items (Next 15-20%)
- Class C: Remaining low-value items

The Classification helps identify which item require strict monitoring and control.

ABC analysis is one of the Standard methods for Inventory Management in Business data management in real-time business data. I used ABC analysis for poultry dataset in the Google Spreadsheets and Python.

A	B	C	D	E
Item_Name	Total_Consumption_Value	Percentage Contribution	Cumulative %	ABC_class
Broiler Feed	50920	35.90%	35.90%	A
Starter Feed	34000	23.97%	59.87%	A
Vaccines(ND)	29400	20.73%	80.60%	B
Vitamins	13125	9.25%	89.85%	C
Disinfectant	8100	5.71%	95.56%	C
Litter Material	6300	.4.44%	100%	C

Fig-5: ABC Classification of Inventory Based on Consumption Value

- The analysis clearly shows that feed-related items consume the highest portion of total inventory cost.
- Broiler feed alone contributes the largest share, making it the most critical cost component in farm operations.
- Starter feed is the second highest expense and directly affects bird growth and productivity.
- Together, Class A items account for the majority of inventory investment and must be purchased carefully to avoid wastage.
- Vaccines under Class B play an important role in bird health and mortality control and should be stocked regularly.
- Class C items contribute less to total cost but are essential for maintaining hygiene and bird immunity.
- Proper classification helps the owner focus more on high-value items rather than monitoring all items equally.
- ABC analysis supports cost reduction by preventing overstocking and expiry-related losses.
- This method improves financial planning and ensures efficient use of working capital in the poultry farm.
- Karthik Reddy Poultry Inventory can managed properly using references from the ABC Classification.

Justification:

Inventory cost and wastage were major concerns in the poultry farm. ABC analysis helps prioritize important inventory items like feed and medicines, ensuring proper stock management and reducing unnecessary expenses. This directly addresses the problem of poor inventory planning and cost inefficiency.

2.6 Justification and Linking of Methods to Business Problems

Regression Analysis → Chick Mortality Problem

Helps identify key factors affecting mortality and supports better temperature control, vaccination, and hygiene practices.

Time Series Analysis → Sales and Revenue Instability

Helps understand demand patterns, sales growth, and revenue fluctuations for better production and financial planning.

ABC Analysis → Inventory Cost and Wastage Problem

Identifies high-cost inventory items requiring strict monitoring and supports efficient stock management.

Descriptive Statistics → Overall Operational Understanding

Provides clear understanding of variability, trends, and performance of mortality, sales, and inventory data.

These analytical methods collectively provide data-driven insights to improve operational efficiency, reduce losses, and increase profitability of the poultry farm.

Vaccination Vs Mortality

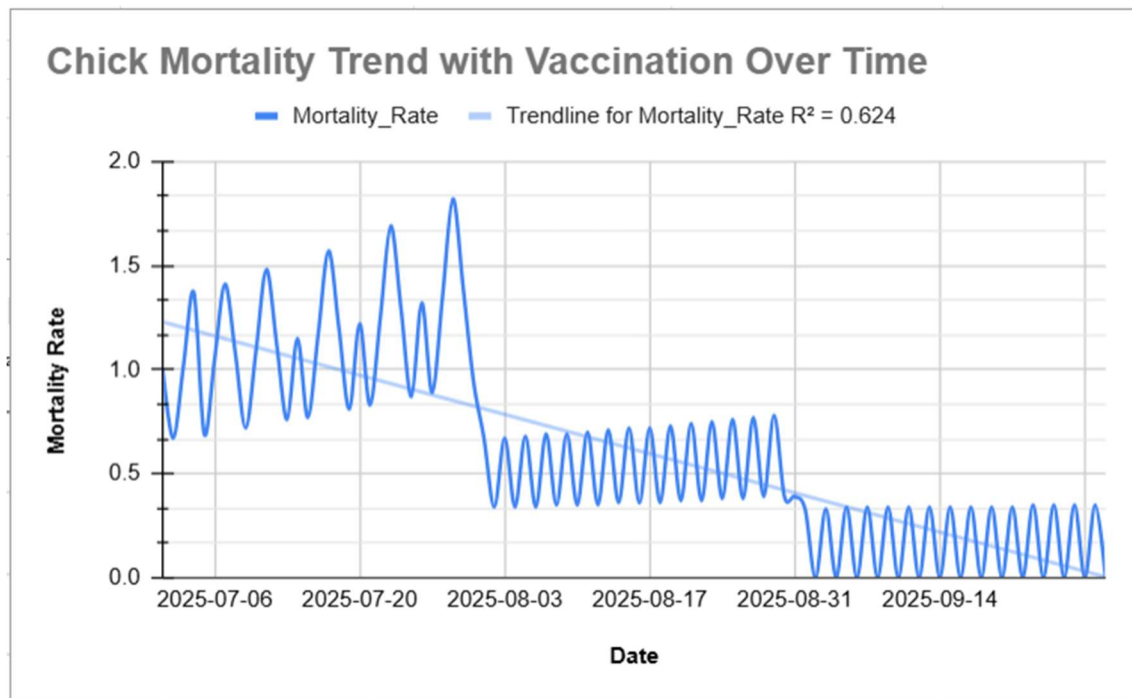


Fig 7: Chick Mortality Trend with Vaccination Over Time

- The graph shows variation in chick mortality rate across different dates.
- Mortality is relatively higher during the initial period of observation.
- A gradual decreasing trend in mortality can be observed over time.
- The downward trendline indicates overall improvement in chick survival.
- On days when vaccination is performed, mortality tends to be lower.
- During non-vaccination periods, mortality is comparatively higher.
- This analysis highlights the role of vaccination along with other factors in improving chick survival.

Mortality vs Litter Condition

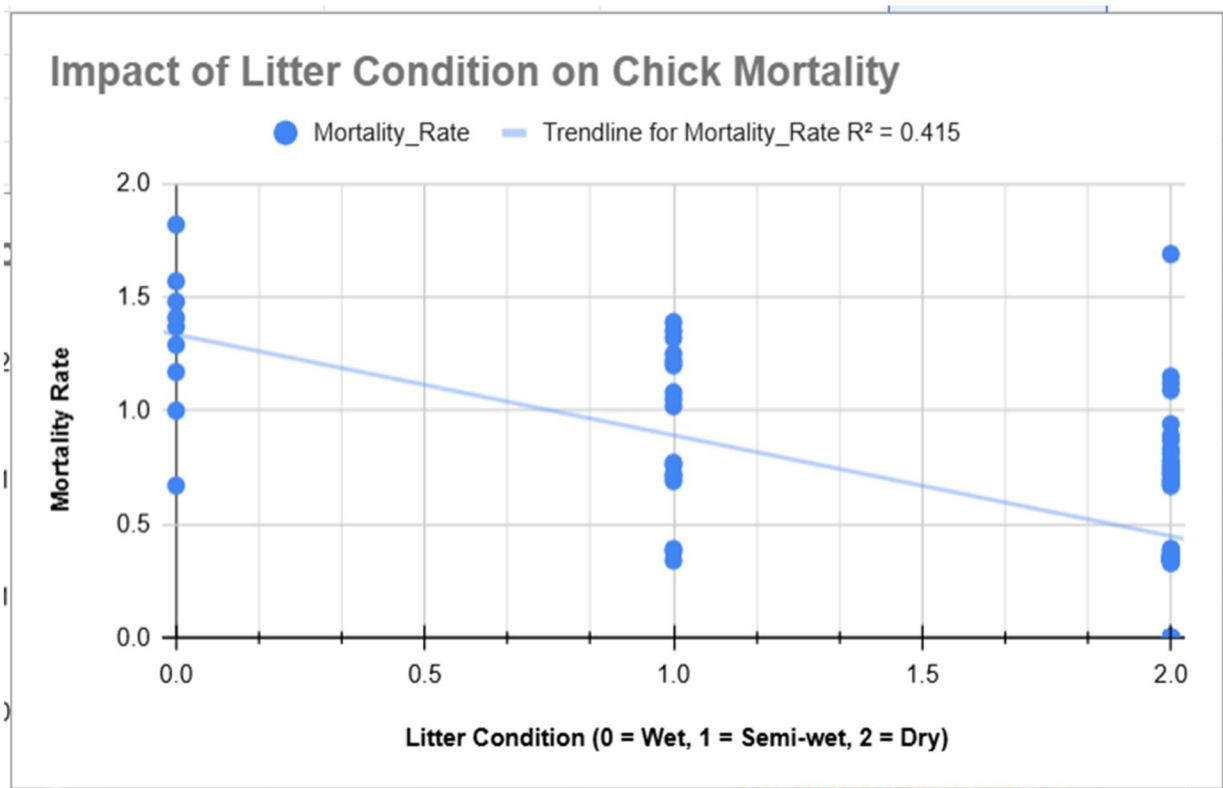


Fig 8: Impact of Litter Condition on Chick Mortality

- The graph shows the relationship between litter condition and chick mortality rate.
- Litter condition is represented on a scale from 0 (wet) to 2 (dry).
- Mortality is higher under wet litter conditions (0) and decreases as litter becomes drier.
- Semi-wet conditions (1) show moderate levels of mortality.
- Dry litter conditions (2) are associated with comparatively lower mortality rates.
- The negative trendline indicates that improved litter conditions reduce mortality.
- The R^2 value (~ 0.415) indicates a moderate relationship between litter condition and mortality.

Combined Results

- The above regression graphs show mortality with each factor separately.
- Correlation values are not very strong, so one factor alone cannot explain mortality clearly.
- So, combined model (multiple regression) is used to understand the overall effect.

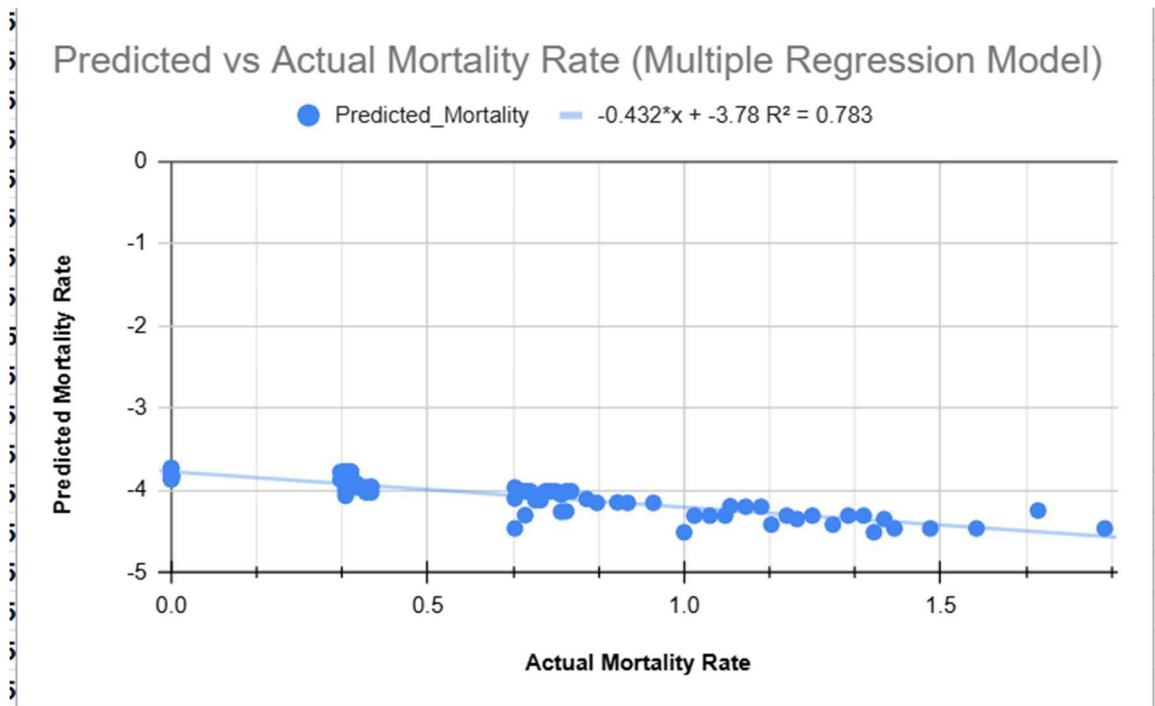


Fig 9: Predicted vs Actual Mortality Rate (Multiple Regression Model)

- The graph compares actual mortality values with predicted values obtained from the regression model.
- The predicted values are calculated using the regression equation, which may produce negative outputs due to the intercept and scaling of variables.
- These negative values do not represent actual mortality, as mortality cannot be negative in real-world conditions.
- The plotted values in the graph are limited to the observed data range and therefore do not extend to extreme values such as -8.63.
- The value (e.g., -8.63) is obtained mathematically from the regression equation for certain inputs but is not present in the dataset used for plotting.
- The number line (axis) reflects only the range of predicted values observed in the dataset and not the full theoretical range of the equation.

- The spread of points indicates that the model captures the general pattern of mortality variation.
- The R^2 value (~ 0.78) indicates that the model explains a significant portion of variation in mortality.
- However, the model has limitations in predicting exact mortality values due to the presence of negative outputs and unobserved factors.
- Overall, the model is useful for understanding relationships among variables rather than making precise mortality predictions.

3.2 Sales Trend Results (Time Series Analysis)

Daily Sales Trend

- Time-Series Analysis helps in identifying sales growth patterns and revenue stability over time.
- The upward trend supports better production planning and demand forecasting for future batches.

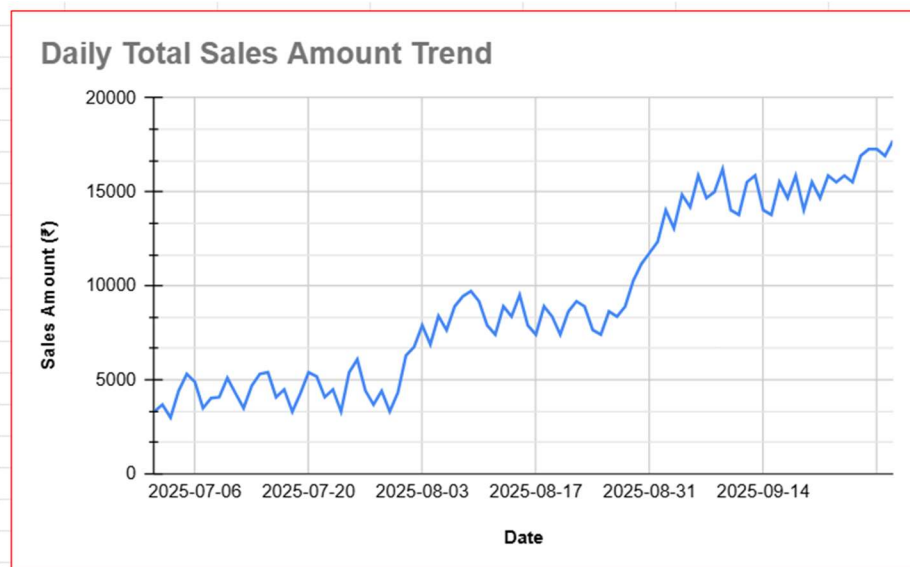


Fig 10: Daily Total Sales amount Trend

- The time-series graph shows a clear upward trend in daily sales from July to September.
- Sales were relatively lower and fluctuating during July, indicating initial instability in demand.
- Sales were relatively lower and fluctuating during July, indicating initial instability in demand.
- September records the highest daily sales values, reflecting strong revenue growth.
- The overall upward movement confirms steady revenue growth during the study period.

Total Sales Amount vs Individual Months

- The time-series graph shows a clear upward trend in daily sales from July to September.
- July has the lowest sales (around 130,000).
- September records the highest sales (around 450,000)
- Due to Dusshera festival, the sales for live chickens increased gradually from August to September.

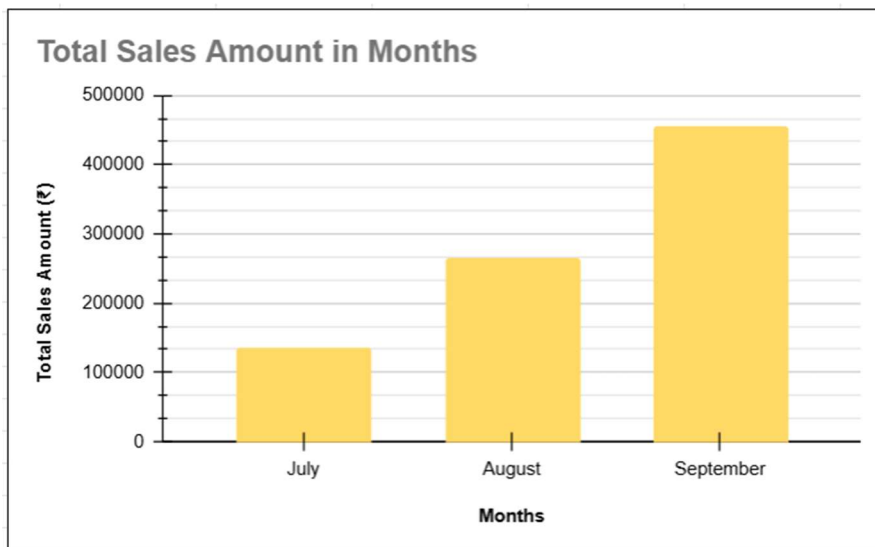


Fig 11: Monthly Sales Amount in poultry farm

3.3 Inventory Classification Results (ABC Analysis)

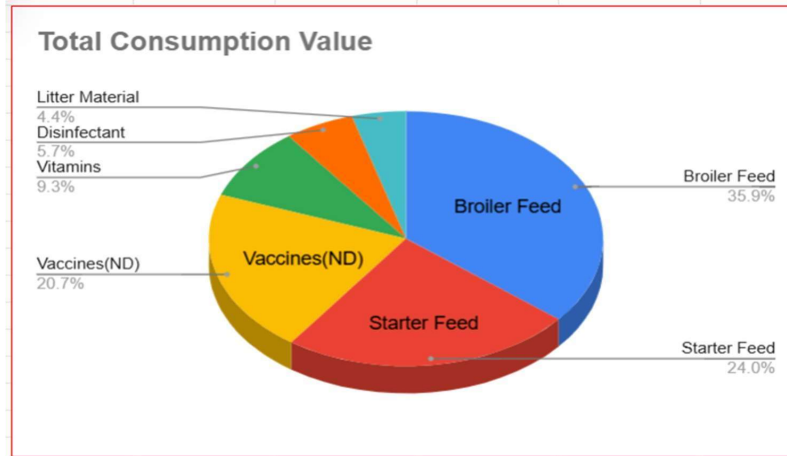


Fig 12: ABC Classification Based on Total Inventory Consumption Value

- Broiler Feed contributes the highest share (around 35–36%), making it the most critical inventory item.
- Starter Feed is the second largest contributor (around 24%), indicating high usage during early bird growth stages.
- Vaccines (ND) account for approximately 20%, showing significant investment in bird health management.
- Vitamins, Disinfectant, and Litter Material together form a smaller percentage, classified under lower consumption categories.



Fig 13: Item-wise Inventory Consumption Value (Descending Order)

- The bar chart represents total consumption value of each inventory item in descending order.
- Broiler Feed shows the highest monetary consumption, confirming its Class A status.
- Starter Feed and Vaccines follow, forming the major portion of inventory expenditure.
- Items like Vitamins, Disinfectant, and Litter Material show comparatively lower consumption values.
- The declining pattern clearly separates high-value, medium-value, and low-value inventory items.
- This visual representation strengthens the ABC classification results numerically derived earlier.

4 Interpretations of Results and Recommendations

4.1 Mortality Reduction Strategy

Interpretation:

The regression results indicates with higher shed Temperature and humidity, with proper ventilation and dry litter condition reduce mortality levels. The model explains significant Portion of mortality variation, This proves that chick death is largely affected by environmental conditions rather and random factors.

Recommendations (SMART):

- Maintain shed temperature within optimal range daily and record it twice per day (Immediate implementation).
- Monitor ventilation conditions daily and correct airflow blockages within 24 hours.
- Ensure litter remains dry by conducting morning and evening inspections (Start immediately).
- For next Batch Cycle, The target is minimum 8-10% reduction in mortality rate in the next production cycle (within 2-3 months).

4.2 Sales Planning and Revenue Growth Strategy

Interpretation:

Sales show an overall increasing trend from July to September, indicating improved demand and revenue. Some daily fluctuations are observed, suggesting the need for better planning.

Recommendations (SMART):

- Monitor daily sales and review monthly performance
- Target ~10% increase in average sales
- Maintain sufficient stock during high-demand periods
- Plan production based on demand trends

4.3 Inventory Cost Optimization Strategy

Interpretation:

Feed items (Broiler and Starter Feed) contribute the highest cost, while medicines and vaccines also have significant impact. Other items have lower cost but remain important.

Recommendations (SMART):

- Monitor high-cost items regularly
- Maintain optimal stock levels
- Check expiry of medicines and vaccines
- Reduce feed wastage

4.4 Implementation Plan

The farm can implement simple monitoring practices such as recording daily temperature and chick mortality. Weekly sales review and monthly inventory checks can help improve planning. Regular vaccination and proper management can reduce mortality and improve overall farm performance